

Arms race co-evolution of *Magnaporthe oryzae* AVR-Pik and rice *Pik* genes driven by their physical interactions

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SUMMARY

Attack and counter-attack impose strong reciprocal selection on pathogens and hosts, leading to development of arms race evolutionary dynamics. Here we show that *Magnaporthe oryzae* avirulence gene AVR-Pik and the cognate rice resistance (*R*) gene *Pik* are highly variable, with multiple alleles in which DNA replacements cause amino acid changes. There is tight recognition specificity of the AVR-Pik alleles by the various *Pik* alleles. We found that AVR-Pik physically binds the N-terminal coiled-coil domain of *Pik* in a yeast two-hybrid assay as well as in an *in planta* co-immunoprecipitation assay. This binding specificity correlates with the recognition specificity between AVR and *R* genes. We propose that AVR-Pik and *Pik* are locked into arms race co-evolution driven by their direct physical interactions.

Keywords: *Magnaporthe oryzae*, *Oryza sativa*, avirulence gene, resistance gene, arms race, coevolution, population genetics.

INTRODUCTION

Antagonistic biotic interactions, including parasitism and predation, impose strong reciprocal selection on living organisms, and are a major driver in evolution. The population geneticist J. B. S. Haldane wrote 'I want to suggest that the struggle against disease, and particularly infectious disease, has been a very important evolutionary agent...' (Haldane, 1949). It is now widely accepted that natural selection imposed by pathogens is one of the major causes of organismal evolution; it has even been proposed that the maintenance of sex in hosts may be attributed to host-pathogen interactions (Ebert and Hamilton, 1996; Morran *et al.*, 2011). Between host and pathogen, two types of evolutionary dynamics have been proposed: (i) arms race and (ii) trench warfare (Woolhouse *et al.*, 2002). In arms race dynamics, a pathogen evolves an allele of a virulence gene that enhances its fitness through suppression of host defense. The frequency of such an allele rapidly increases in the pathogen population, due to the fitness advantage it confers to the carrier pathogen, eventually replacing the older allele. In turn, a new allele of a host counter-defense gene that nullifies the effect of the pathogen virulence gene emerges, and its allele frequency increases rapidly to become fixed in the host population. These reciprocal

processes are repeated over a long time. In trench warfare dynamics (Stahl *et al.*, 1999), also called Red Queen dynamics (Woolhouse *et al.*, 2002), negative frequency-dependent selection causes the allele frequencies of the corresponding genes of pathogen and host to oscillate over time, resulting in maintenance of the same set of alleles in the population for a longer time. The two types of dynamics leave contrasting patterns of DNA signatures on the genes of involved species: genes involved in arms race dynamics experience recurrent selective sweeps and exhibit rapid evolution, resulting in a larger number of inter-species amino acid replacements and low levels of polymorphism within the species, whereas genes involved in trench warfare co-evolution show a larger amount of within-species polymorphisms. These two models may be applied at the population level as well as the species level, provided that populations are effectively isolated from each other. Population-level analysis appears to be more relevant to pathogen-crop interactions (Terauchi and Yoshida, 2010).

The existence of gene-for-gene relationships between host resistance (*R*) and pathogen avirulence (*AVR*) genes between plants and their pathogens is well established (Flor, 1956; Dangl and Jones, 2001). However, studies on

co-evolution of *R* and *AVR* genes are still limited. A DNA polymorphism study of the *Arabidopsis thaliana* *Rpm1* gene (Stahl *et al.*, 1999) showed that the gene contains two allelic classes (an *R* class conferring resistance and an *S* class that does not) that diverged in ancient past, suggestive of long-term balancing selection presumably caused by the trench warfare type of interaction. On the other hand, the *RPP13* *R* gene of *Arabidopsis thaliana* was shown to harbor an extreme level of amino acid polymorphisms (Rose *et al.*, 2004). This corresponds to the large amount of non-synonymous polymorphisms in *ATR13*, a cognate *AVR* gene of *RPP13* in *Hyaloperonospora parasitica* (Allen *et al.*, 2004). These large amounts of non-synonymous polymorphisms of host and pathogen are thought to be maintained by reciprocal selection at the population level by arms race dynamics. A study of flax (*Linum usitatissimum*) and flax rust (*Melampsora lini*) interactions is the most thoroughly studied case of arms race co-evolution in plant–pathogen interactions. The flax *L* locus, encoding a nucleotide binding site/leucine-rich repeat (NBS-LRR) protein, harbors 12 allelic variants with extensive sequence diversity (Ellis *et al.*, 1999). *Melampsora lini* *AvrL567* is an avirulence gene recognized by the *L5*, *L6* and *L7* alleles of the flax *L* gene (Dodds *et al.*, 2004). *AvrL567* also has 12 allelic variants in six strains, and its diversity is extreme, with more non-synonymous DNA sequence polymorphisms than synonymous ones. Dodds *et al.* (2006) showed that *L* protein and *AvrL567* protein physically interact in yeast two-hybrid analysis, and the strength of physical binding recapitulates the interaction specificity observed *in planta*, indicating that *L* and *AvrL567* are evolving by arms race dynamics through physical interactions at the population level. More recently, the same group analyzed DNA polymorphisms of *M. lini* *AvrP123* and *AvrP4* avirulence genes, and suggested that their polymorphisms are maintained by both arms race and trench warfare dynamics (Barrett *et al.*, 2009).

Rice blast caused by the ascomycete fungus *Magnaporthe oryzae* is the most devastating disease of rice worldwide. Control of blast disease has mainly relied on deployment of resistant (*R*) genes of rice. So far more than 40 *R* genes have been named, and ten have been cloned (Ballini *et al.*, 2008; Liu *et al.*, 2010; Terauchi *et al.*, 2011). On the other hand, only six *AVR* genes of *M. oryzae* have been cloned to date. *ACE1* is an unusual avirulence gene as it encodes an enzyme that is involved in synthesis of a secondary metabolite (Böhnert *et al.*, 2004). *AvrPita* and *AvrPizt* have been cloned by map-based cloning (Orbach *et al.*, 2000; Li *et al.*, 2009). Using an association genetics approach, we previously cloned three *AVR* genes, *AVR-Pik/km/kp*, *AVR-Pia* and *AVR-Pii* (Yoshida *et al.*, 2009). A DNA polymorphism study of these *AVRs* in rice-infecting field strains showed very contrasting profiles. *AVR-Pia* and *AVR-Pii* showed only presence/absence polymorphisms without any nucleotide changes (Yoshida *et al.*, 2009), while *AVR-Pizt* had a trans-

poson inserted in the promoter region and a single point mutation (Li *et al.*, 2009). *AVR-Pita* showed very diverse alleles, with complete or partial deletions, transposon insertions, point mutations and even translocations (Orbach *et al.*, 2000; Zhou *et al.*, 2007; Dai *et al.*, 2010; Takahashi *et al.*, 2010; Chuma *et al.*, 2011). Finally, *AVR-Pik/km/kp* exhibited extensive nucleotide changes leading to amino acid replacements in addition to presence/absence polymorphisms (Yoshida *et al.*, 2009). The rice *R* gene *Pik* has been reported to show multiple alleles (*Pik*, *Pikm*, *Pikp*, *Piks* and *Pikh*) with differential resistance to *M. oryzae* races. Hereafter, to avoid confusion in nomenclature, we use '*Pik*' to indicate the locus name, and '*Pik**' to represent the conventional *Pik* allele. The rice *R* gene allele *Pikm*, the *AVR-Pikm* cognate, has been cloned by Ashikawa *et al.* (2008) and Zhai *et al.* (2011). *Pikm* comprises two protein-coding ORFs, *Pikm-1* and *Pikm-2*, separated by a 2.5 kb non-coding region and oriented in opposite directions. In this paper, we refer to the two ORFs of the *Pik* gene as *Pik-1* and *Pik-2*. Constanzo and Jia (2010) studied DNA sequence polymorphisms of the *Pik* locus for *Pik-1* and *Pik-2* using 15 rice cultivars. They found a large amount of DNA polymorphisms clustered in the *Pik-1* region corresponding to the coiled-coil (CC) domain, whereas only a low level of DNA polymorphisms were observed in *Pik-2*. Furthermore, they found two amino acid residues at positions 229 and 252 in the CC domain that showed extensive polymorphisms, and proposed that these amino acid residues may be used as diagnostic markers for the classification of *Pik* alleles: *Pik**, *Pikm*, *Pikp*, *Piks* and *Pikh*.

We are interested in the co-evolutionary dynamics of *M. oryzae* *AVR* genes and rice *R* genes. Here we show that *M. oryzae* *AVR-Pik* alleles and rice *Pik* alleles exhibit high levels of amino acid changes. We show that a subset of combinations of *AVR-Pik* and the CC domain of *Pik-1* physically bind, and this binding specificity mainly determines the recognition specificity of *AVR-Pik* alleles by *Pik* alleles. We propose that this physical interaction results in arms race co-evolution between *AVR-Pik* and *Pik*.

RESULTS

DNA polymorphisms in *M. oryzae* *AVR-Pik* and rice *Pik*

M. oryzae *AVR-Pik* is a 113 amino acid protein with a 21 amino acid signal peptide (Figure 1a) (Yoshida *et al.*, 2009). Five alleles of *AVR-Pik* (*AVR-Pik-A*, *B*, *C*, *D* and *E*) were identified in 21 isolates of *M. oryzae* from Japan. The *AVR-Pik* DNA sequence is highly variable with a nucleotide diversity (Nei, 1987) of 7.1×10^{-3} , which is two orders higher than the mean value for the entire genome (8.2×10^{-5}) as revealed by EcoTILLING (Yoshida *et al.*, 2009). The five *AVR-Pik* alleles are different from each other by a total of five DNA substitutions, all of which cause amino acid changes (Figure 1b and Figure S1). We found a paralog of

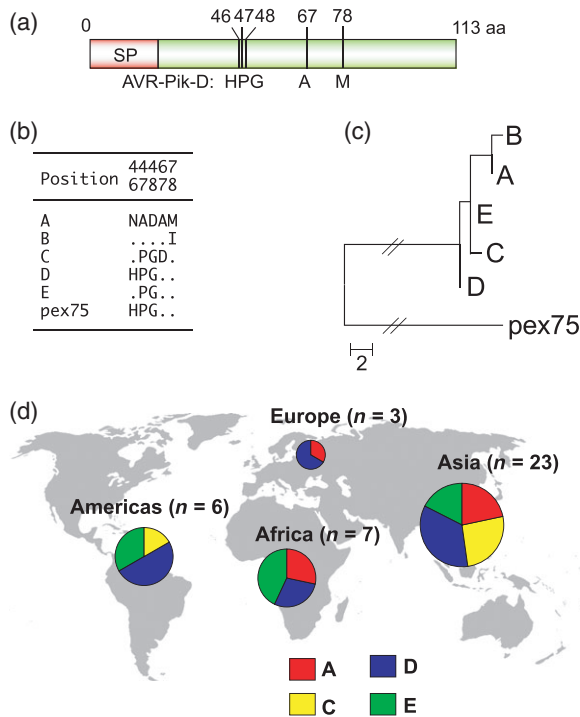


Figure 1. Genetic diversity of *Magnaporthe oryzae* AVR-Pik.

(a) Structure of the AVR-Pik protein. SP indicates signal peptide. The positions of variable residues are indicated above, and the amino acids of the AVR-Pik-D protein for the variable positions are given below. (b) Amino acids of five AVR-Pik allelic products (A, B, C, D and E) as well as pex75 for the five variable sites (46, 47, 48, 67 and 78). (c) Phylogenetic tree for the five AVR-Pik allelic products generated by the maximum parsimony method. The pex75 sequence was used to root the tree. Branch length corresponds to mutation number (see the scale below the tree). (d) World distribution of AVR-Pik-A, -C, -D and -E alleles.

AVR-Pik, pex75, in the genome of *M. oryzae* (Figure S1, and see Supplementary Dataset 2 in Yoshida *et al.*, 2009). AVR-Pik-D shared identical amino acid residues with pex75 for the five positions that segregate among the AVR-Pik alleles (Figure 1b). Phylogenetic analysis of the AVR-Pik amino acid polymorphisms using the pex75 sequence as an out-group (Nei and Kumar, 2000) revealed the consensus maximum-parsimony tree (Figure 1c). This tree suggests that the AVR-Pik-D allele is most likely the ancestral allele, from which the AVR-Pik-E, -C, -A and -B alleles were derived.

To assess the worldwide distribution of AVR-Pik alleles in rice-infecting *M. oryzae*, we performed PCR amplification and DNA sequencing of AVR-Pik genes from 39 isolates collected from Asia ($n = 23$), Africa ($n = 7$), Europe ($n = 3$) and America ($n = 6$) (Figure 1d and Table S1). These isolates were chosen to be representative of *M. oryzae* genetic diversity (Tharreau *et al.*, 2009; Saleh *et al.*, 2012). AVR-Pik-B, which was found in an isolate from Japan (isolate 9505-3; Yoshida *et al.*, 2009) was not identified in the 39 isolates.

AVR-Pik-D was the most frequent allele (15 of 39). The three other alleles showed similar frequencies (7–9 of 39). The AVR-Pik-A, -C, -D and -E alleles were distributed worldwide, with the few exceptions probably being explained by differences in sample size between continents.

The rice *Pik* gene comprises two adjacent inversely oriented ORFs encoding NBS-LRR proteins (*Pik-1* and *Pik-2*) (Figure 2) (Ashikawa *et al.*, 2008). DNA polymorphisms of *Pik-1* and *Pik-2* were studied for the C-terminal LRR region by Ashikawa *et al.* (2010) and for the N-terminal CC domain region by Constanzo and Jia (2010). The latter authors identified two residues (229 and 252) in the N-terminal CC region of *Pik-1* that are highly variable and useful for diagnosis of *Pik* alleles (Constanzo and Jia, 2010). This observation suggests a possibility that this region may be involved in recognition of AVR-Pik alleles. To assess the levels of DNA polymorphisms of *Pik* corresponding to the N-terminal CC domain region using a larger number of rice accessions, we performed a DNA sequencing analysis of an 800 bp region of *Pik-1* and *Pik-2* for a total of 37 *Oryza* accessions collected from around the world, comprising 33 accessions of *O. sativa* and four accessions of *Oryza rufipogon* (the wild progenitor species of *O. sativa*) (Table S2). The *O. sativa* accessions used were randomly selected from a worldwide rice collection (Kojima *et al.*, 2005) that covers *O. sativa* genetic diversity. Phylogenetic analysis of the aligned sequences (Figure S2) revealed that there are two major clades (Group 1 and 2) in both *Pik-1* and *Pik-2*. These clades do not correspond to taxonomic separation of the studied *O. sativa* accessions into ssp. *japonica* and ssp. *indica*. There is a perfect linkage disequilibrium between *Pik-1* and *Pik-2*: all the varieties carrying an allelic variant of *Pik-1* belonging to Group 1 carry an allelic variant of *Pik-2* belonging to Group 1, and the same for Group 2 allelic variants (Figure 2). The well-characterized *Pik* alleles (*Pik**, *Pikm*, *Pikp*, *Piks* and *Pikh*) belong to Group 1 for *Pik-1* and *Pik-2*. Amino acid sequence comparison of the CC domain (Figure S2) confirmed that the two sites (229 and 252) are segregating for three or four amino acid residues (Q, E or K at position 229; P, H, D or R at position 252), as reported by Constanzo and Jia (2010). If we assume that the *Pik* allelic state of rice accessions can be predicted by the two diagnostic sites, the largest number of studied rice accessions harbor *Piks* (13 accessions), followed by *Pikp* (six accessions), *Pik** (three accessions), *Pikm* (three accessions) and *Pikh* (one accession) (Figure 2). Within Group 1 of *Pik-1*, there are four clades with high bootstrap values: a clade for *Piks*, *Pikh* and *Pikm*, one for *Pikp*, one for Kanto51-type *Pik** and one for Toto-type *Pik** (Figure 2). For both *Pik-1* and *Pik-2*, the divergence between Group 1 and 2 accounts for the largest part of DNA variation for both synonymous and non-synonymous sites (Table 1). *Pik-1* contains substantial DNA polymorphisms within Group 1 ($\pi_{\text{syn}} = 0.046$; $\pi_{\text{non-syn}} = 0.035$) and Group 2 ($\pi_{\text{syn}} = 0.013$; $\pi_{\text{non-syn}} = 0.033$),

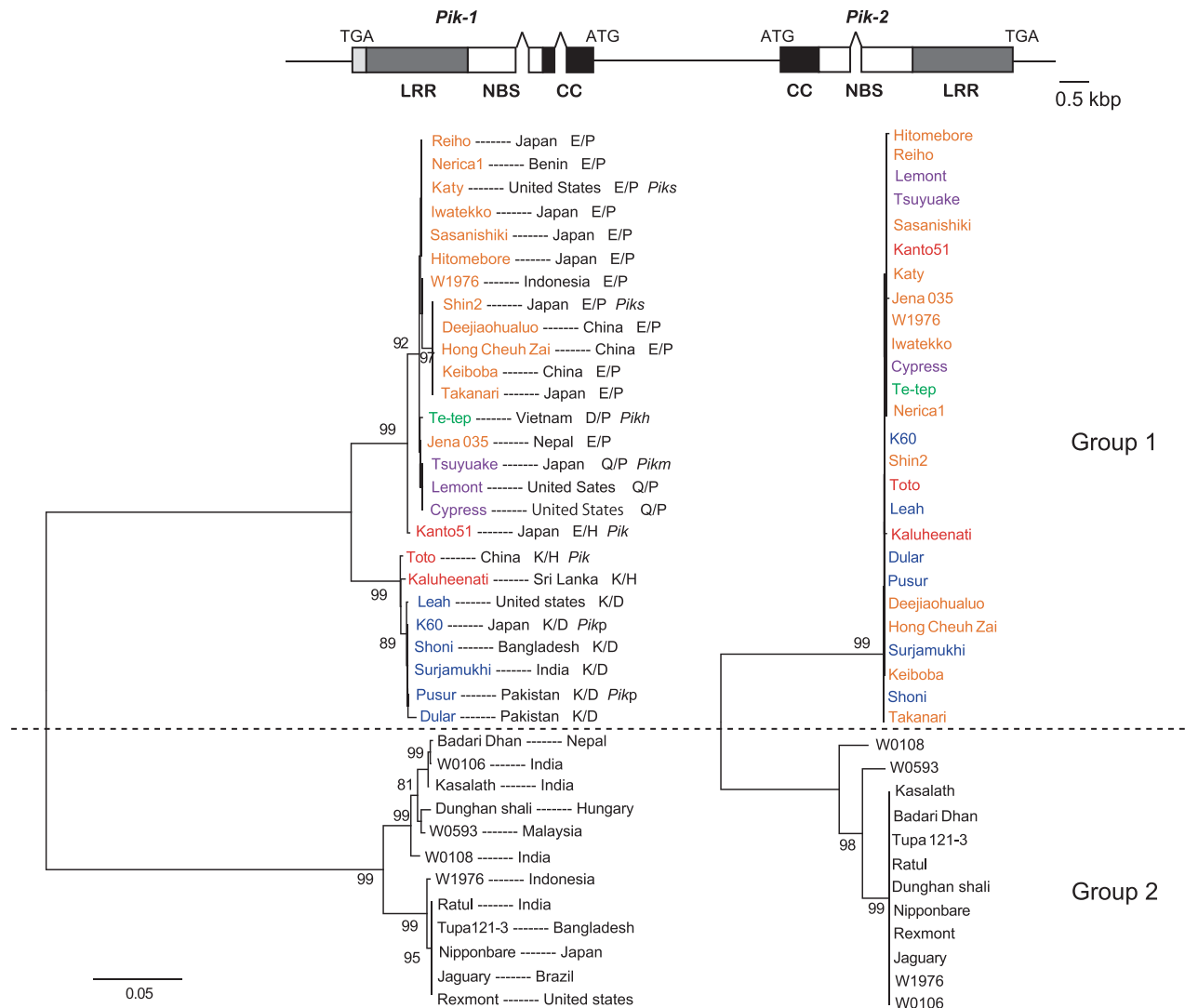


Figure 2. Genetic diversity of the DNA regions corresponding to the coiled-coil (CC) domain of *Pik-1* and *Pik-2* genes.

Structure of the *Pik-1* and *Pik-2* genes (top). CC, coiled-coil domain; NBS, nucleotide-binding site; LRR, leucine-rich repeats. Neighbor-joining trees based on DNA sequences of 0.8 kbp regions corresponding to the CC domains of *Pik-1* and *Pik-2* genes. DNA sequences of *O. sativa* (33 accessions) and *O. rufipogon* (four accessions) were used (see Table S2 for details). Following the cultivar names/accession numbers, the places of origin, and the amino acids for the two highly variable residues (positions 229 and 252) proposed to be useful for *Pik* allele diagnosis (Constanzo and Jia, 2010) are given. Based on the amino acids in the two residues, putative *Pik* allele names were predicted and are indicated by different colors: *Pik** (red), *Pikp* (blue), *Pikh* (green), *Piks* (orange) and *Pikm* (purple). The cultivars are divided into two groups: Group 1 and 2. A bar below the tree indicates genetic distance estimated as described by Jukes and Cantor (1969). Bootstrap values exceeding 80% after 1000 re-samplings are shown on the branches.

but *Pik-2* lacks polymorphisms in Group 1 ($\pi_{\text{syn}} = 0$; $\pi_{\text{non-syn}} = 0.001$). Tajima's test of neutral mutation hypothesis applied to all the samples for *Pik-1* and *Pik-2* showed statistically significant ($P < 0.05$) positive values for Tajima's D (Tajima, 1989) (Table 1). This result suggests either that balancing selection is imposed on this locus, or recent merging of two divergent populations. As there is no indication of recent merging of *O. sativa* populations, we hypothesize that this positive Tajima's D is probably caused by balancing selection imposed on this locus or the region linked to this locus.

AVR-Pik alleles are specifically recognized by different *Pik* alleles

Maintenance of multiple alleles of AVR-Pik in *M. oryzae* and *Pik* in rice populations prompted us to study whether each allele of AVR-Pik is recognized by specific *Pik* alleles. To address this possibility, we created isogenic lines of *M. oryzae* harboring different AVR-Pik alleles. The *M. oryzae* isolate Sasa2 does not harbor AVR-Pik, so it can infect rice cultivars with any *Pik* alleles. We transformed Sasa2 individually with constructs harboring four alleles of AVR-Pik:

Table 1 Summary of nucleotide polymorphisms for the regions encoding the CC domain of *Pik-1* and *Pik-2* in *Oryza sativa* and *Oryza rufipogon*

Population	Number of samples	<i>Pik-1</i>						<i>Pik-2</i>					
		Synonymous sites			Non-synonymous sites			Synonymous sites			Non-synonymous sites		
		Size	π	<i>D</i>	Size	π	<i>D</i>	Size	π	<i>D</i>	Size	π	<i>D</i>
All	38	164.26	0.182	2.75**	528.74	0.158	2.84**	168.93	0.126	2.08*	590.07	0.062	2.25*
Group 1	26	188.14	0.046	1.92 [†]	582.86	0.035	1.95 [†]	177.31	0	NA	614.69	0.001	−0.76 NS
Group 2	12	178.04	0.013	2.28*	604.96	0.033	1.26 NS	169.13	0.021	−1.26 NS	592.87	0.010	−1.68 [†]
<i>O. sativa</i>	33	169.67	0.158	1.89 [†]	538.33	0.140	1.90 [†]	175.29	0.107	1.98 [†]	613.71	0.057	1.87 [†]
Group 1	25	188.15	0.047	1.98 [†]	582.85	0.036	2.04*	177.31	0	NA	614.69	0.001	−0.76 NS
Group 2	8	182.52	0.013	1.70 NS	615.48	0.029	1.39 NS	178.83	0	NA	634.17	0	NA
<i>O. rufipogon</i> ^a	5	158.73	0.155	−1.29***	537.27	0.147	−1.28***	168.83	0.144	−0.74 NS	590.17	0.069	−0.71 NS
Group 2	4	177.08	0.012	0.18 NS	605.92	0.035	0.53 NS	173.38	0.055	0.62 NS	600.63	0.028	0.19 NS

D: Tajima's *D*.
[†]*P* < 0.10, **P* < 0.05, ***P* < 0.01, ****P* < 0.001; NS, not significant; NA, cannot be applied because of no polymorphic site.
^a *O. rufipogon* accession W1976 was heterozygous for Group1 and Group2 allelic variants of *Pik*, so that the total number of samples (sequences) was 5.

AVR-Pik-D, *-E*, *-A* and *-C*. As *AVR-Pik-B* occurred only in one isolate from Japan, we did not include it in the analysis. All alleles were driven by the native promoter of the *AVR-Pik-D* allele. Each transgenic *M. oryzae* line was used for inoculation of four rice cultivars harboring different alleles of *Pik* (Nipponbare, *Pik*[−]; K60, *Pikp*; Kanto51, *Pik*^{*}; Tsuyuake, *Pikm*) (Figure 3). Nipponbare lacks known *Pik* alleles; however, it contains a *Pik* orthologous sequence, thus the *Pik* ortholog of this cultivar is indicated as *Pik*[−]. As expected, the cultivar Nipponbare without a known *Pik* allele (*Pik*[−]) was suscepti-

ble to all four isolates harboring the *AVR-Pik-D*, *-E*, *-A* or *-C* alleles, respectively. Cultivar K60 containing the *Pikp* *R* gene was resistant to *M. oryzae* harboring *AVR-Pik-D* but susceptible to *M. oryzae* harboring *AVR-Pik-E*, *-A* or *-C*. Cultivar Kanto51 containing the *Pik*^{*} *R* gene was resistant to *M. oryzae* harboring *AVR-Pik-D* or *-E* transgenes, but susceptible to *M. oryzae* harboring *AVR-Pik-A* or *-C*. It is notable that the reaction of *Pik*^{*} to *AVR-Pik-A* was not clear cut: disease lesions develop, but the speed of lesion development was slower than in other compatible interactions, and

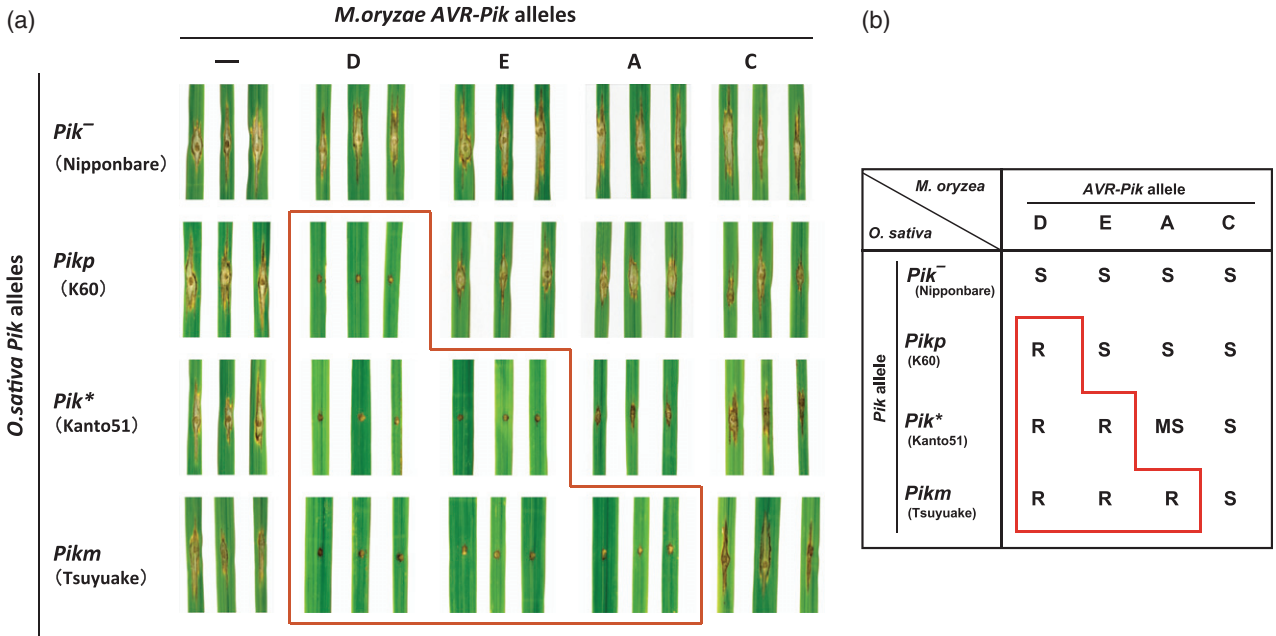


Figure 3. Recognition specificity of four *Magnaporthe oryzae* *AVR-Pik* alleles by rice *Pik* alleles. (a) *M. oryzae* isolate Sasa2 without any *AVR-Pik* alleles was transformed with empty vector (—), *AVR-Pik-D* (D), *AVR-Pik-E* (E), *AVR-Pik-A* (A) or *AVR-Pik-C* (C), and punch-inoculated into rice cultivars Nipponbare (*Pik*[−]), K60 (*Pikp*), Kanto51 (*Pik*^{*}) and Tsuyuake (*Pikm*). *AVR-Pik/Pik* combinations resulting in incompatibility are outlined in red. (b) Simplified diagram showing the interactions observed in (a). R, plant showed resistance; S, plant showed susceptibility; MS, plant showed moderate susceptibility.

thus this reaction is indicated as moderate susceptibility (Figure 3). Finally, cultivar Tsuyuake containing the *Pikm* *R* gene was resistant to *M. oryzae* harboring *AVR-Pik-D*, *-E* or *-A*, but susceptible to that harboring *AVR-Pik-C*. These results suggest that the three rice *Pik* alleles *Pikp*, *Pik** and *Pikm* have differential recognition specificity towards the four *M. oryzae* alleles (*AVR-Pik-D*, *-E*, *-A* and *-C*) that differ from each other by a maximum of four amino acid changes (Figure 1b). We also tested the recognition specificity of two other known *Pik* alleles, *Piks* and *Pikh*, against the four *AVR-Pik* alleles (Figure S3). The recognition specificities of *Piks* and *Pikp* were identical, and those of *Pikh* and *Pikm* were identical. Therefore, for the rest of study we focus only on the three *Pik* alleles *Pikp*, *Pik** and *Pikm* and the *Pik* ortholog in Nipponbare (*Pik*⁺). Amino acid sequence comparison of the CC domain revealed that *Pikh* and *Pikm* are very similar, whereas *Piks* and *Pikp* are distantly related. We attempted to find common amino acid residues shared only by *Piks* and *Pikp*, but could not identify any such residues. At present, it is difficult to infer how *Piks* and *Pikp* share the same recognition specificity (Figure S2).

Recognition specificity of *AVR-Pik* alleles by *Pik* alleles is determined by strength of the hypersensitive response (HR) cell death

Our blast inoculation assay suggested that *M. oryzae* *AVR-Pik* alleles are specifically recognized by different *Pik* alleles. To assess whether this recognition involves hypersensitive response (HR) cell death, we performed rice cell death assays by transient over-expression of *M. oryzae* *AVR-Pik* alleles in rice protoplasts by transfection using electroporation (Figure 4a) (Yoshida *et al.*, 2009). In this assay, we monitored the viability of rice cells by the activity of firefly luciferase, the gene for which is transfected into rice protoplasts together with *AVR-Pik*. To drive expression of *AVR-Pik* alleles and the luciferase gene, we used the strong maize ubiquitin promoter (Christensen and Quail, 1996). If the rice cells harbor an *R* gene cognate of the *AVR-Pik* allele leading to HR cell death, we see a reduction in luciferase activity. Luciferase activity was monitored by measuring the luminescence 40 h after transfection, by adding luciferin and ATP to the cell extracts (Figure 4a). No HR cell death was observed in Nipponbare rice cells (*Pik*⁺) without a known *Pik* allele after transfection with all four *AVR-Pik* alleles: *AVR-Pik-D*, *-E*, *-A* and *-C* (Figure 4b). In cultivar K60 containing *Pikp*, HR cell death occurred only when *AVR-Pik-D* was transfected, in agreement with the results of the blast inoculation assay (Figure 3). In cultivar Kanto51 containing *Pik**, *AVR-Pik-D* caused HR cell death but *AVR-Pik-C* did not, consistent with the results of the inoculation assay (Figure 3). Transfection of Kanto51 (*Pik**) with two other alleles, *AVR-Pik-E* and *-A*, resulted in an intermediate level of HR cell death (Figure 4b), although only the former interaction supported incompatibility in the inoculation assay (Figure 3). The HR

spectrum of the cultivar Tsuyuake with *Pikm* is similar to that of Kanto51 (*Pik**): *AVR-Pik-D* triggered strong HR, *AVR-Pik-C* caused no significant HR, and *AVR-Pik-E* and *-A* resulted in an intermediate HR (Figure 4b). However, *AVR-Pik-D*, *-E* and *-A* all caused incompatibility in the inoculation assay (Figure 3). Therefore, there are variable cases in which an intermediate level of the HR supports incompatibility (*Pik**/*AVR-Pik-E*, *Pikm*/*AVR-Pik-E* and *Pikm*/*AVR-Pik-A*) or does not (*Pik**/*AVR-Pik-A*). Apart from these cases, the results of *M. oryzae* inoculation assay and protoplast HR assay conformed with each other, suggesting that incompatibility is mainly determined by the HR reaction.

AVR-Pik-D and the CC domain of *Pik-1* physically interact

AVR-Pik has five alleles that differ from each other by DNA changes corresponding to one to four amino acid replacements (Figure 1b). *Pik* has five named alleles (*Pik**, *Pikp*, *Pikm*, *Piks* and *Pikh*) that differ by amino acid replacements in the CC domain (Figures 2 and S2), and can be diagnosed by the amino acids in the two sites 229 and 252 (Constanzo and Jia, 2010). Amino acid changes in *AVR-Pik* determine the recognition by different rice *Pik* alleles (Figures 3 and 4). The observed high levels of amino acid polymorphisms and the recognition specificity between *AVR* and *R* can most easily be explained if *AVR* and *R* proteins physically interact. To test this possibility, we performed a yeast two-hybrid assay to study direct binding between *AVR-Pik* and *Pik*.

As *AVR-Pik-D* is recognized by *Pikm* (Figures 3 and 4), consisting of the two ORFs *Pikm-1* and *Pikm-2*, we first tested physical interactions between *AVR-Pik-D* and *Pikm-1* and between *AVR-Pik-D* and *Pikm-2*. As shown in Figure 5(a), we observed interaction between full-length *Pikm-1* and *AVR-Pik-D* when the former was used as bait and the latter as prey, but no interaction was detected between full-length *Pikm-2* and *AVR-Pik-D*. Next we used the CC domain of *Pikm-1* to test the interaction with *AVR-Pik-D*, and found that it indeed interacts with *AVR-Pik-D* (Figure 5a). If we swapped bait and prey, we did not observe binding between full-length *Pikm-1* and *AVR-Pik-D* (Figure 5b). However, the CC domain of *Pikm-1* used as prey interacted with *AVR-Pik-D* used as bait (Figure 5b), suggesting that *AVR-Pik-D* does indeed physically bind the CC domain of *Pikm-1*. The LRR (*Pikm-1*-LRR) and NBS (*Pikm-1*-NBS) regions did not interact with *AVR-Pik-D* (Figure 5b). Expression of all bait and prey proteins was confirmed by Western blot analysis (Figure S4).

To test whether the molecular interaction between *Pikm-1* and *AVR-Pik-D* is observed *in planta*, we performed a co-immunoprecipitation assay in *Nicotiana benthamiana* leaves. The full-length *Pikm-1* protein, and the CC domain only and the NBS-LRR domain only of *Pikm-1* were tagged with the 3xFLAG epitope, resulting in 3xFL-*Pikm-1*-full, 3xFL-*Pikm-1*-CC, and 3xFL-*Pikm-1*-NBS-LRR, respectively,

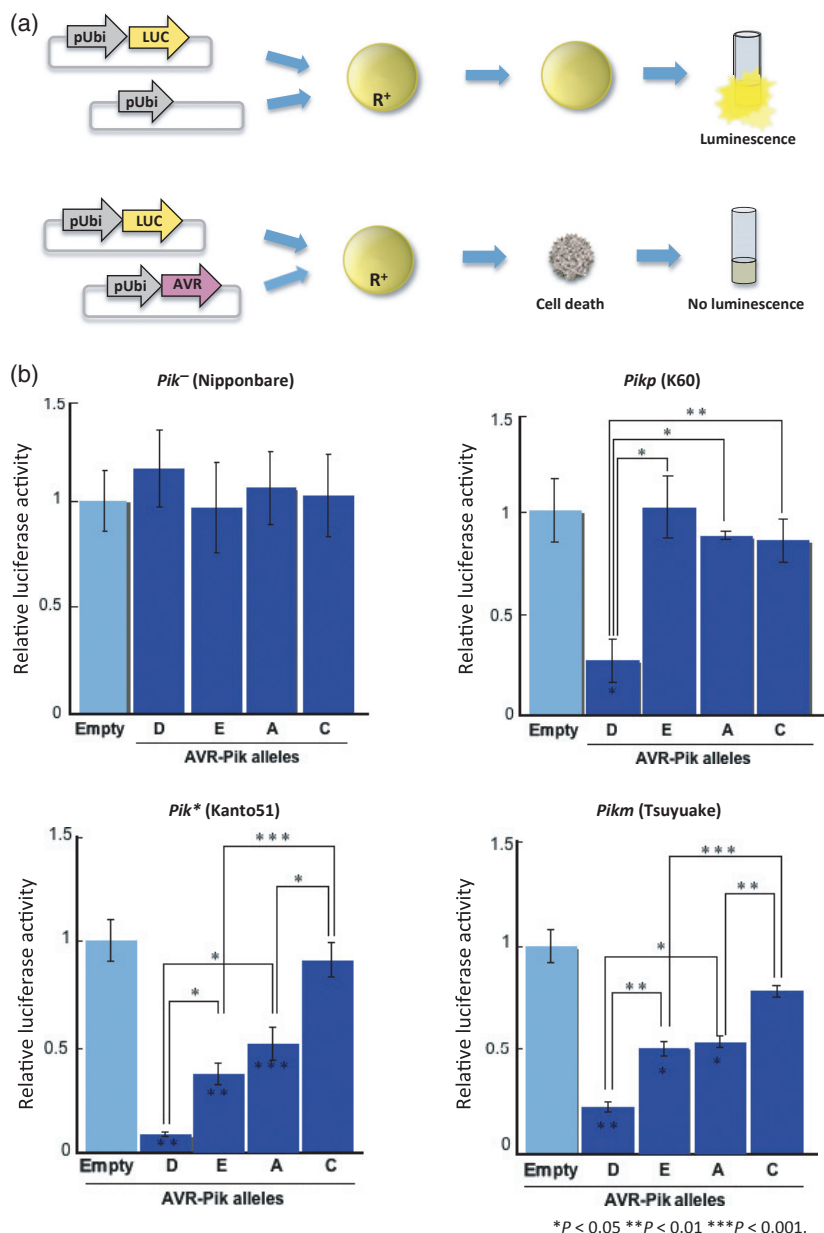


Figure 4. Hypersensitive response (HR) cell death assay for AVR-Pik/Pik interactions.

(a) Simplified scheme of the transient HR cell death assay. A plasmid containing the firefly luciferase gene (*LUC*) driven by a strong maize ubiquitin promoter (*pUbi*) was transfected to rice protoplasts by electroporation. Cell viability was monitored by the luciferase activity (top). If an AVR gene driven by *pUbi* is co-transfected with *LUC* into protoplasts of a rice cultivar harboring the cognate *R* gene, the AVR-*R* interaction causes HR cell death, resulting in a reduction in luciferase activity (bottom).

(b) Viability of rice cells as judged by luciferase activity after transfection with plasmids harboring different AVR-Pik alleles (D, AVR-Pik-D; E, AVR-Pik-E; A, AVR-Pik-A; C, AVR-Pik-C). Luciferase activity is indicated as relative activity, with the value for the empty vector set as 1. Standard errors are indicated. Statistical significance of differences is indicated by asterisks: **P* < 0.05; ***P* < 0.01; ****P* < 0.001.

and AVR-Pik-D was tagged with the HA (haemagglutinin) epitope to generate AVR-Pik-D-HA. These proteins were transiently over-expressed in *N. benthamiana* leaves by agroinfiltration (Win *et al.*, 2011). As shown in Figure 5(c), all proteins were successfully expressed *in planta*. When immunoprecipitation of AVR-Pik-D-HA was performed using anti-HA antibody, 3xFL-Pikm-1-full and 3xFL-Pikm-1-CC were co-immunoprecipitated as detected by the anti-FLAG antibody, but 3xFL-Pikm-1-NBS-LRR was not (Figure 5c). These observations confirm the result obtained by the yeast two-hybrid assay, and demonstrate that the CC domain of Pikm-1 and AVR-Pik-D physically interact *in planta*.

Binding specificity between Pik-1 and AVR-Pik determines the recognition specificity between *Pik* and AVR-Pik

Next, we tested binding specificity between the full-length products of three *Pik*-1 alleles (*Pikp*-1, *Pik*^{*}-1 and *Pikm*-1) and the Nipponbare *Pik*-1 ortholog *Pik*⁻, and the protein products of four AVR-Pik alleles (AVR-Pik-D, -E, -A and -C) using a yeast two-hybrid assay (Figure 6a). Expression of all bait and prey proteins was confirmed by Western blot analysis (Figure S4). In the case of *Pik*⁻-1 from Nipponbare, α -galactosidase activity was observed even in the absence of prey protein. This result suggests that the *Pik*⁻-1 bait plasmid has auto-activation ability in yeast cells, so we were unable to

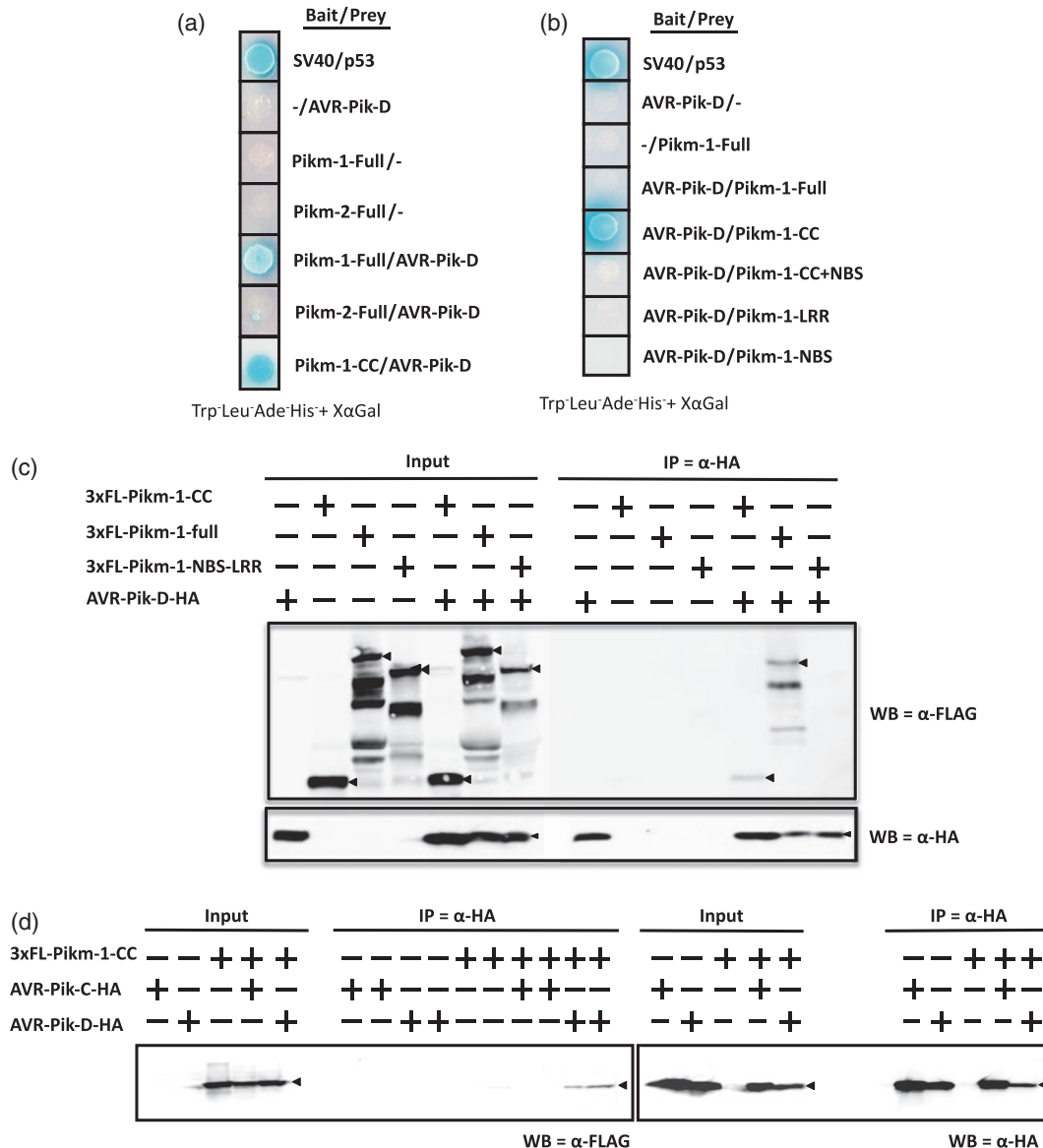


Figure 5. Physical interactions of AVR-Pik and Pik proteins.

(a) Yeast two-hybrid experiment results between AVR-Pik-D and Pikm-1 or Pikm-2, using R as bait and AVR as prey. SV40/p53, positive control for interaction; -, empty vector used; Full, full-length protein; CC, coiled-coil region.

(b) Similar to (a), but with bait and prey swapped. NBS, nucleotide binding sites; LRR, leucine-rich repeats.

(c) Results of an *in planta* co-immunoprecipitation assay after over-expression of the CC domain of Pikm-1 (3xFL-Pikm-1-CC), full-length Pikm-1 (3xFL-Pikm-1-full) or the NBS-LRR domain of Pikm-1 (3xFL-Pikm-1-NBS-LRR) and AVR-Pik (AVR-Pik-D-HA) in *Nicotiana benthamiana* leaves by agroinfiltration. Input: the protein was detected by anti-FLAG antibody (α-FLAG: top) or by anti-haemagglutinin (HA) antibody (α-HA: bottom) (left). IP = α-HA: after immunoprecipitation with anti-HA antibody, co-immunoprecipitated protein was detected using α-FLAG (top) or α-HA (bottom) antibody (right).

(d) Results of *in planta* co-immunoprecipitation assay after over-expression of the CC domain of Pikm-1 (3xFL-Pikm-1-CC) and AVR-Pik-C or AVR-Pik-D in *N. benthamiana* leaves. 3xFL-Pikm-1-CC co-immunoprecipitated with AVR-Pik-D-HA, but not with AVR-Pik-C-HA.

assess its interaction with AVR-Pik allelic products. Pikp-1, Pik*-1 and Pikm-1 all lack this auto-activation ability. We found a specific interaction between Pikp-1 and AVR-Pik-D, between Pik*-1 and AVR-Pik-D and -E, and between Pikm-1 and AVR-Pik-D, -E and -A. Neither Pikp-1, Pik*-1 or Pikm-1 bound AVR-Pik-C (Figure 6a). As far as Pikp-1, Pik*-1 and Pikm-1 are concerned, their binding specificities with AVR-

Pik allelic products perfectly correspond to the recognition specificities of *Pikp*, *Pik** and *Pikm* towards AVR-Pik alleles (Figure 3).

We further tested the interactions of the CC domain of Pik-1 allelic variants with AVR-Pik variants (Figure 6b,c). Expression of all bait and prey proteins was confirmed by Western blot analysis (Figure S4). Again, the CC domain of Pik-1

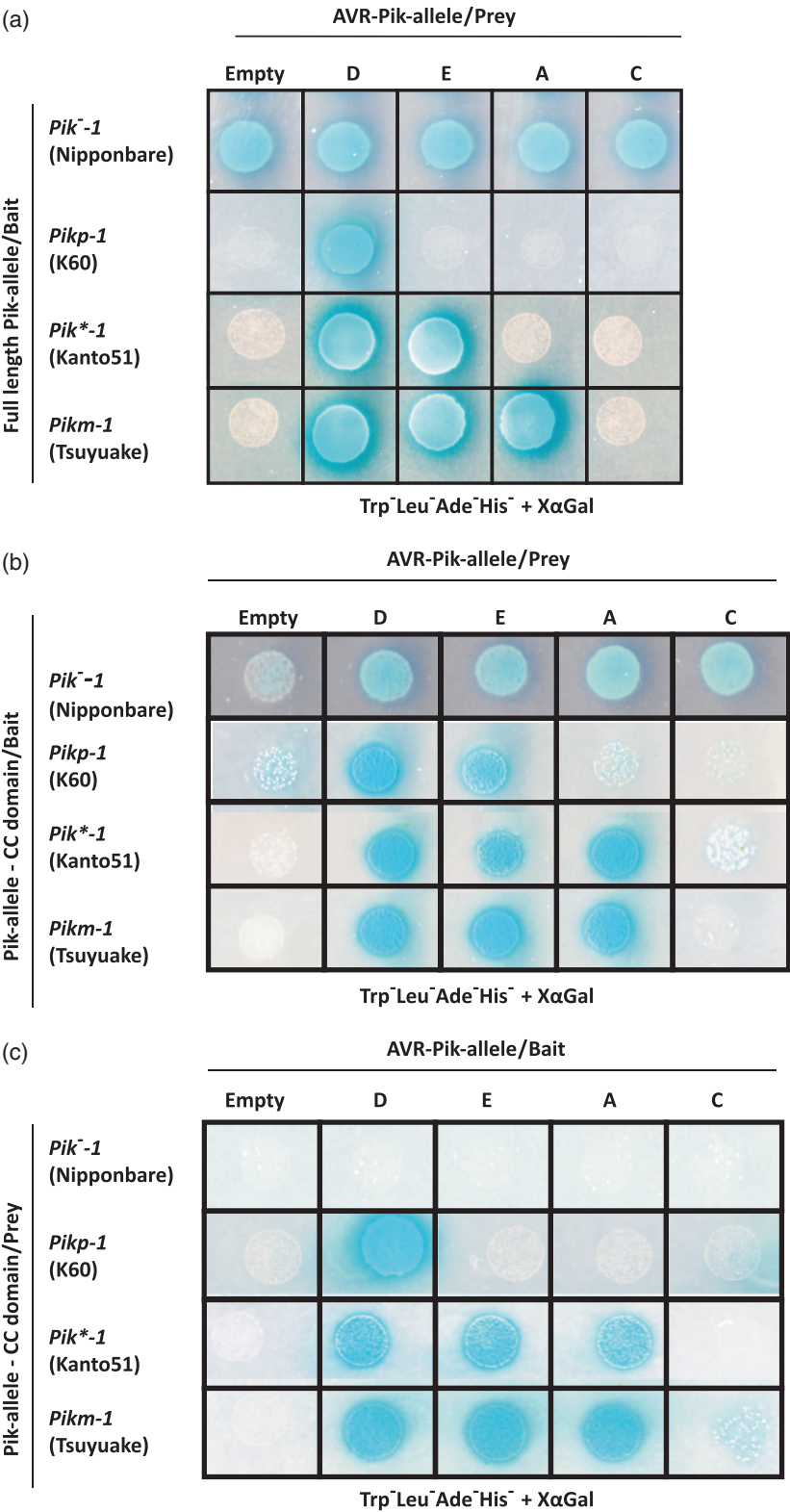


Figure 6. Binding specificity between AVR-Pik and Pik allelic products as revealed by the yeast two-hybrid assay. (a) Yeast two-hybrid experiment results for interactions between the full-length proteins of Pik-1 alleles (*Pikp-1*, *Pik*-1* and *Pikm-1*) as well as the Pik-1 ortholog in Nipponbare (*Pik⁻-1*) used as bait, and AVR-Pik proteins used as prey: AVR-Pik-D (D), AVR-Pik-E (E), AVR-Pik-A (A) and AVR-Pik-C (C). (b) As in (a), but testing interactions between the CC domain of Pik-1 allelic products used as bait and AVR-Pik proteins as prey. (c) Same as B, but bait and prey were swapped.

showed auto-activation. The interaction patterns of the CC domain of Pik-1 variants with AVR-Pik variants were identical to those observed when the full-length Pik-1 was used

(Figure 6a), except for binding of the *Pik*-1* CC domain with AVR-Pik-A. The same result was obtained after swapping bait and prey (Figure 6c). The fact that the *Pik*-1* CC domain

binds AVR-Pik-A whereas the Pik^{*}-1 full-length protein does not, suggests that it is not only the CC domain that determines the binding specificity between Pik-1 and AVR-Pik allelic products.

We further tested the specificity of binding *in planta* by co-immunoprecipitation assays (Figure 5d). The CC domain of Pikm-1 co-immunoprecipitated with AVR-Pik-D but not with AVR-Pik-C, consistent with the result obtained in the yeast two-hybrid assay (Figure 6).

AVR-Pik-A expression levels affect its recognition by Pik^{*}

In the inoculation assay with *M. oryzae* harboring different alleles of AVR-Pik as transgenes, we observed that the interaction between Pik^{*} and AVR-Pik-A resulted in reduced development of disease lesions, which is not a typical compatibility (Figures 3 and 7a). By contrast, when we inoculated natural isolates Kyu92-22 and ARC-P90-19A that contain AVR-Pik-A into Kanto51 containing Pik^{*} allele, clear-cut compatibility was observed (Figure 7a). To determine the reason of this discrepancy, we studied the expression

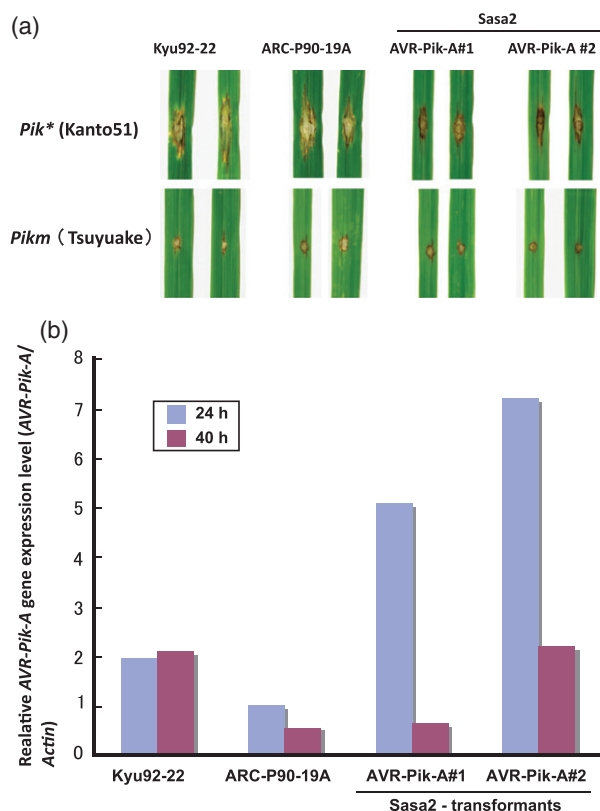


Figure 7. Gene expression levels of *Magnaporthe oryzae* AVR-Pik-A affect its recognition by the rice Pik^{*} R gene.

(a) Leaf punch inoculation results for two wild-type *M. oryzae* isolates (Kyu92-22 and ARC-P90-19A) and two transgenic *M. oryzae* Sasa2 isolates expressing AVR-Pik-A (AVR-Pik-A#1 and AVR-Pik-A#2) inoculated into Kanto51 expressing Pik^{*} or Tsuyuake expressing Pikm.

(b) Relative expression levels of AVR-Pik-A mRNA in the two wild-type isolates and two transgenic isolates 24 and 40 h after inoculation.

levels of AVR-Pik-A in two transgenic isolates harboring AVR-Pik-A (#1 and #2) as well as the two natural isolates (Figure 7b). We found consistently higher levels of expression of AVR-Pik-A in the transgenic lines (#1 and #2) compared to the natural isolates (Kyu92-22 and ARC-P90-19A) 24 h after inoculation, suggesting that the expression levels of AVR-Pik-A may affect recognition by Pik^{*}: the higher the level of AVR-Pik-A expression, the higher the chance of recognition by Pik^{*}.

DISCUSSION

Direct interaction between AVR-Pik and Pik

Our results showed that *M. oryzae* AVR-Pik-D and rice Pikm-1 proteins directly interact in a yeast two-hybrid assay (Figure 5a) and an *in planta* co-immunoprecipitation assay (Figure 5c). This interaction is mediated by the Pikm-1 CC domain (Figure 5a,b). There is binding specificity between different AVR-Pik allelic products and different Pik-1 allelic products (Figure 6). This binding specificity determines the recognition specificity of Pik alleles towards different AVR-Pik alleles (Figure 3) mediated by HR induction (Figure 4). Direct binding between fungal AVR and plant R proteins was first reported for *M. oryzae* AVR-Pita and rice Pita proteins (Jia *et al.*, 2000). In this case, single amino acid substitutions in the Pita LRR region or the AVR-Pita protease motif resulting in disruption of the physical interaction caused loss of resistance in the plant. Our finding is also similar to that observed in interactions between flax rust AVR-L567 alleles and flax L alleles (Dodds *et al.*, 2006), in which 12 alleles of AVR-L567 are differentially recognized by L alleles by direct physical interactions of their products. To our knowledge, the present study is the third example of direct AVR-R interactions in fungal-plant host interactions.

It is interesting to note that AVR-Pik/Pik interactions involve the CC domain, but not the LRR domain, of the R protein. The majority of previous studies suggest that the LRR domain of R proteins is involved in AVR recognition (powdery mildew *AvrMla1* recognition by barley *Mla1*, Shen *et al.*, 2003; PVX recognition by potato *Rx*, Rairdan and Moffett, 2006) and in physical interaction with AVR proteins (blast AVR-Pita/rice Pita, Jia *et al.*, 2000; *Hyaloperonospora arabidopsidis* ATR1/Arabidopsis RPP1, Krasileva *et al.*, 2010; rust AVR-L567/flax L, Ravensdale *et al.*, 2011). Recently, Chen *et al.* (2012) reported that the potato R protein RB binds to *Phytophthora infestans* AVR protein IPI-O through its CC domain. Together with the results of Chen *et al.* (2012), the CC domain-mediated interactions between R and AVR proteins described here reveal a new aspect of AVR/R interactions.

Co-evolution of AVR-Pik and Pik

Phylogenetic analysis suggests that AVR-Pik-D is the ancestral allele among the five AVR-Pik alleles (Figure 1c).

AVR-Pik-D is also the allele recognized by the largest number of *Pik* alleles (*Pik**, *Pikp*, *Pikm*, *Piks* and *Pikh*) as tested in the current study (Figures 3 and S3). The AVR-Pik-E allele appears to be derived from AVR-Pik-D by one amino acid change (Figure 1b). AVR-Pik-E is recognized by three *Pik* alleles (*Pik**, *Pikm* and *Pikh*). There appears to be a tendency for the older AVR-Pik alleles to be recognized by a larger number of *Pik* alleles. The largest number of studied rice accessions harbor *Piks* (13 accessions), followed by *Pikp* (six accessions) (Figure 2). It is interesting to note that both *Piks* and *Pikp* can recognize AVR-Pik-D but not AVR-Pik-E, AVR-Pik-A or AVR-Pik-C. The most widespread rice *Pik* alleles recognize the oldest AVR allele, AVR-Pik-D. Based on these findings, we propose that the observed variation patterns may be interpreted as the result of co-evolution between the *M. oryzae* AVR-Pik gene and rice *Pik* gene (Figure 8). For instance, the following co-evolution scenario may be possible: first the AVR-Pik-D allele evolved from the common ancestor of AVR-Pik and *pex75*. Rice cultivars that recognize AVR-Pik-D may have had a selective advantage, so an *R* gene such as *Pikp* and *Piks* may have been selected. This selection may have resulted in evolution of AVR-Pik-E, which evades recognition by *Pikp* and *Piks*. In response to this situation, humans deployed another *Pik* allele, *Pik**, that can recognize both AVR-Pik-D and AVR-Pik-E. Under this selection pressure, *M. oryzae* evolved the AVR-Pik-A allele by three amino acid changes from AVR-Pik-E. Again, another *Pik* allele, *Pikm* (or *Pikh*), was deployed that recognizes AVR-Pik-D, -E and -A.

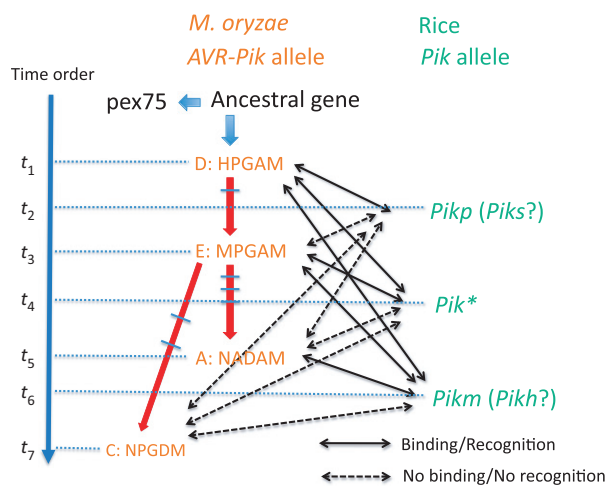


Figure 8. Possible scenario for *Magnaporthe oryzae* AVR-Pik- rice *Pik* co-evolution.

Chronological order is given on the left (time order). AVR-Pik homolog *pex75* and AVR-Pik-D were derived from an ancestral *M. oryzae* gene. AVR-Pik-D is recognized by *Pikp* and *Piks*, thus the altered allele AVR-Pik-E evolved. In response to this situation, another *Pik* allele, *Pik**, evolved that can recognize both AVR-Pik-D and -E. Then, yet another AVR-Pik allele, AVR-Pik-A, was derived that cannot be recognized by *Pikp* and *Pik**. Next, the rice *R* gene *Pikm* (or *Pikh*) was utilized that recognizes AVR-Pik-D, -E and -A. Then another AVR-Pik allele, AVR-Pik-C, evolved that cannot be recognized by any of the three *Pik* alleles. Binding to AVR-Pik allelic products was not tested for *Piks* and *Pikh*.

Under this selection pressure, *M. oryzae* then evolved the AVR-Pik-C allele that evades recognition by the three *Pik* alleles tested in this study. As the majority of the wild rice (*O. rufipogon*) studied belongs to Group 2 with no capacity to recognize any of AVR-Pik alleles, we hypothesize that AVR-Pik/*Pik* co-evolution occurred after domestication of rice. Also, the lack of synonymous substitutions among AVR-Pik alleles indicates that AVR-Pik allele diversification occurred in the very recent past, presumably as a result of serial deployment of different *Pik* alleles by humans. Whether the worldwide distribution of each AVR-Pik allele results from multiple independent events of selection or from a unique selection event followed by migration remains to be tested.

Dose effect of recognition of AVR-Pik-A by *Pik**

The CC domain of *Pik*-1* binds AVR-Pik-A (Figure 6b,c), but the full-length *Pik* protein cannot bind AVR-Pik-A (Figure 6a), resulting in no recognition of AVR-Pik-A by the *Pik** allele under natural conditions (Figure 7a). These results suggest that other regions of the *Pik** protein interfere with binding between the *Pik** CC domain and AVR-Pik-A. However, if AVR-Pik-A is over-expressed as a transgene, the protein level becomes higher than the wild-type allele, and thus the isolates are partially recognized by *Pik** (Figure 7a). This suggests a threshold effect of recognition. If the abundance of AVR is below a certain threshold, the number of cases in which R-AVR binding occurs is too low to trigger a sufficient level of HR to cause incompatibility. If the number of cases of R-AVR binding is augmented by increasing the amount of AVR and/or R, this may cause an HR that is strong enough to make the interaction incompatible. Such a dosage effect suggests that the expression levels of AVR and R are under strong natural selection. It will be interesting to systematically examine the expression levels of proteins involved in pathogen-host interactions.

Comparison of interactions between AVR-Pia and Pia and between AVR-Pik and *Pik*

In previous studies, it was shown that DNA polymorphisms of most *M. oryzae* avirulence genes cloned to date include both presence/absence and nucleotide substitutions (Orbach *et al.*, 2000; Zhou *et al.*, 2007; Li *et al.*, 2009; Yoshida *et al.*, 2009; Dai *et al.*, 2010; Takahashi *et al.*, 2010; Chuma *et al.*, 2011). In contrast, two of them, AVR-Pia and AVR-Pii, showed only presence/absence polymorphisms. The easiest way for *M. oryzae* to avoid recognition by rice *R* genes appears to be to lose the AVR gene. The variation in AVR-Pia and AVR-Pii appear to fit this manner in avoiding recognition by cognate rice *R* genes. However, in the case of AVR-Pik, *M. oryzae* contains multiple alleles with amino acid substitutions. This suggests that loss of AVR-Pik may have a fitness penalty that is larger than that for AVR-Pia and AVR-Pii. To understand the fitness con-

tribution of each AVR effector in the absence of a cognate *R* gene, identification of host proteins targeted by each effector is important.

EXPERIMENTAL PROCEDURES

DNA sequence analysis

Genomic DNA of all the plant materials was extracted from fresh leaves using a DNeasy plant mini kit (Qiagen, <http://www.qiagen.com/>). A fragment containing the region encoding the CC domain of *Pik-1* or *Pik-2* was amplified from genomic DNA using PrimeSTAR GXL polymerase (Takara, <http://www.takara-bio.com/>). The primers for *Pik-1* alleles belonging to Group 1 were 5'-ATGGAGGCGGCTGCCATG-3' and 5'-GCTTGAGCGAAAATGTCTGC-3'. The primers for *Pik-1* alleles belonging to Group 2 were 5'-ATGGCGTATACAGCGTCGC-3' and 5'-TCGGTGAGATTGGACCCGGA-3'. The primers for *Pik-2* alleles belonging to Group 1 were 5'-ATGGAGTTGGTGATAGTG-3' and 5'-GATCATCGTCCGTTGTTCTGC-3'. The primers for *Pik-2* alleles belonging to Group 2 were 5'-ATGGAGTTGGCGGTAGTGCTTC-3' and 5'-TCATCAATCAAGAGGATATA-3'. Sequence reactions were performed using BigDye Terminator version 1.1 (Applied Biosystems, <http://www.appliedbiosystems.com/>). An Applied Biosystems 3130xl genetic analyzer was used to determine sequences. Sequencing primers were designed at approximately 500 bp intervals. We used the nucleotide sequences of AVR-*Pik-A* (AB498876), -*B* (AB498877), -*C* (AB498878), -*D* (AB498875) and -*E* (AB498879) to estimate nucleotide diversity and construct the tree.

Alignments of the sequences were performed using ClustalW (Thompson *et al.*, 1994) and manually adjusted based on visual inspection to avoid misalignment. The DNASP program version 5.0 (Librado and Rozas, 2009) was used for the population genetic analysis. After indel removal, the nucleotide diversity π (Nei, 1987) was estimated. To investigate departures from neutrality, Tajima's (1989) test was performed. Phylogenetic trees for *Pik-1*, *Pik-2* and AVR-*Pik* were constructed using the MEGA program version 5 (Tamura *et al.*, 2011).

M. oryzae materials

To create isogenic lines of *M. oryzae* harboring different AVR-*Pik* alleles, we transformed Sasa2 individually with expression vectors for four alleles of AVR-*Pik*: AVR-*Pik-D*, -*E*, -*A* and -*C*. We used Sasa2 harboring AVR-*Pik-D* as utilized by Yoshida *et al.* (2009). For construction of the AVR-*Pik-A*, -*C* or -*E* expression vectors pCB1531:AVR-*Pik* promoter:AVR-*Pik-A*, -*C* or -*E*, a 0.3 kb fragment containing an AVR-*Pik* allele was amplified using the primers described by Yoshida *et al.* (2009), digested with *Xba*I and *Bam*HI, and replaced the ORF of AVR-*Pik-D* of pCB1531:AVR-*Pia* promoter:AVR-*Pik-D* (Yoshida *et al.*, 2009), generating pCB1531:AVR-*Pia* promoter:AVR-*Pik-A*, -*C* or -*E*. A 2.2 kb fragment containing the AVR-*Pik* promoter was amplified using primers NotI-pex31-U1 (Yoshida *et al.*, 2009) and 5'-GCTCTAGACAAATAATGTCTTTGCAACAAAG-3', digested using *Not*I and *Xba*I, and replaced the AVR-*Pia* promoter of pCB1531:AVR-*Pia* promoter:AVR-*Pik-A*, -*C* or -*E*, generating pCB1531:AVR-*Pik* promoter:AVR-*Pik-A*, -*C* or -*E*. DNAs from the Ina168, 70-15 and P2 isolates were used as PCR templates for the AVR-*Pik-A*, -*C* and -*E* alleles, respectively.

Pathogenicity assay

Rice leaf blade spot inoculation (see Kanzaki *et al.*, 2002) was performed with *M. oryzae* strains. Disease lesions were photographed 10 days after inoculation. Rice seedlings (cvs Nipponbare, K60, Kanto51 or Tsuyuake) at the fourth leaf stage were used for inoculation. Rice seeds of *Piks* (Shin2 monogenic lines) or *Pikh* (K3

monogenic lines) were provided by the International Rice Research Institute (<http://www.irri.org/>).

Cell viability assay in rice protoplasts

Protoplasts were isolated from rice etiolated seedlings (cvs Nipponbare, K60, Kanto51 or Tsuyuake) as described by Yoshida *et al.* (2009). PCR-amplified cDNAs of four alleles of AVR-*Pik* (AVR-*Pik-D*, -*A*, -*E* and -*C*) were digested with *Bam*HI and inserted into *Bam*HI sites of the pAHC17 vector under the control of the maize ubiquitin promoter (Christensen and Quail, 1996). To monitor cell viability of protoplasts, the firefly luciferase gene (*LUC*) under the control of the maize ubiquitin promoter was used as a reporter gene. Alleles of AVR-*Pik* were co-transfected together with *LUC* plasmids into rice protoplasts by electroporation. Luciferase activity as an index of cell viability was measured 40 h after transfection using a luciferase assay system (Promega, <http://www.promega.com/>), and the reduction in luminescence was compared with a control comprising empty vector-transfected protoplasts.

Yeast two-hybrid assay

Signal peptide-truncated cDNA fragments amplified from each AVR-*Pik* allele were inserted into *Eco*RI and *Bam*HI sites of pGADT7 (prey) or pGBKT7 (bait) vectors (Clontech, <http://www.clontech.com/>). PCR-amplified fragments of the *Pik-1* allele or the *Pik-2* allele were isolated from seedlings of various rice cultivars (cvs Nipponbare, K60, Kanto51 or Tsuyuake) using PrimeSTAR GXL DNA polymerase (Takara), and cloned into pGADT7 or pGBKT7 using primer sets and restriction enzymes as listed in Table S3. The various combinations of bait and prey were transformed into yeast strain AH109 using the polyethylene glycol/lithium acetate (PEG/LiAc) method. Protein-protein interactions between bait and prey were evaluated by blue coloration and growth of yeast on basal medium lacking Trp, Leu, Ade and His but containing 5-Bromo-4-Chloro-3-indolyl α -D-galactopyranoside (X- α -gal) (Clontech). Details of plasmids used are given in Table S3.

In planta co-immunoprecipitation assay

cDNA fragments of the AVR-*Pik-C* or -*D* allele fused with the HA epitope tag were inserted into the *Bam*HI and *Pst*I sites of the pCambia1300S binary vector (Cambia, <http://www.cambia.org/>). PCR-amplified cDNAs corresponding to the CC domain region of Tsuyuake *Pikm1* were fused to the triple FLAG epitope tag and inserted into the *Spe*I and *Pst*I sites of the pCambia1300S binary vector. These binary vectors were introduced into *Agrobacterium* strain GV3101. *Agrobacterium* cultures harboring binary vectors were infiltrated into leaves of *Nicotiana benthamiana*. Leaf samples taken 5 days after inoculation were ground in liquid nitrogen and solubilized in extraction buffer containing 10 mM dithiothreitol, 2% polyvinylpolypyrrolidone and proteinase inhibitor cocktail (Sigma-Aldrich, <http://www.sigmaaldrich.com/>). The extracts were immunoprecipitated using anti HA-conjugated agarose beads (Sigma-Aldrich) and eluted using HA peptide (Sigma-Aldrich) as described by Win *et al.* (2011). Immunocomplexes were labeled using anti-FLAG peroxidase-conjugated antibody or anti-HA peroxidase-conjugated antibody (Sigma-Aldrich), and detected using an ECL-advance detection kit (GE Healthcare, <http://www3.gehealthcare.com/>). Details of plasmids used are given in Table S3.

Quantitative RT-PCR

Leaf sheaths of rice cv. Moukoto at the four- to five-leaf stage were inoculated with a suspension of *M. grisea* isolate Sasa2 (Yoshida *et al.*, 2009). Total RNA was isolated from leaf sheaths 24 or 40 h after inoculation using an RNeasy plant mini kit (Qiagen). Single-

strand cDNA was synthesized from DNase-treated RNA using oligo(dT) nucleotides and ReverTra Ace (Toyobo, <http://www.toyobo-global.com/>). Quantification of gene expression was performed using a Step OnePlus real-time PCR system (Applied Biosystems) according to the manufacturer's instructions. Levels of *AVR-Pik-A* mRNA as studied by cDNA PCR amplification by primer pair: (5'-ACGTCAAGATGCTGGAACCTGT-3' and 5'-CAACCTGGAGGGAAGTCGCC-3') were measured using the constitutively expressed actin gene as studied by cDNA PCR amplification by primer pair: (5'-GCCGTCTCCCGTCCATT-3' and 5'-CTGGCCCATACCAATCATGAT-3') as a control for normalization.

ACKNOWLEDGEMENTS

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of this article:

Figure S1. Amino acid sequence alignment of AVR-Pik-A, -B, -C, -D and -E and their ortholog pex75.

Figure S2. Amino acid sequence alignment of the Pik-1 CC region.

Figure S3. Rice leaf punch inoculation results for AVR-Pik alleles inoculated into rice lines harboring *Piks* and *Pikh*.

Figure S4. Confirmation of expression of bait and prey proteins in the yeast two-hybrid assay by Western blot analysis.

Table S1. Geographical distribution of AVR-Pik alleles throughout the world.

Table S2. Rice plant materials used for DNA sequencing analysis of *Pik* genes.

Table S3. Plasmids and primers used in the cell viability assay, the yeast two-hybrid assay or the *in planta* co-immunoprecipitation assay.

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